

Anuj Patil

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Summary

Final-year Computer Engineering student who builds and ships. Designed a live static analysis tool (CodeEcoScan) that quantifies the carbon footprint of Python code, built autonomous AI agents for real-time misinformation detection, and architected RAG-based document systems with vectorized PostgreSQL, Redis caching, and Docker microservices. Comfortable owning full-stack projects end-to-end – from Spring Boot and FastAPI backends to React frontends – with hands-on experience in LLM fine-tuning (LoRA), LangChain pipelines, and production deployments on Railway and Render.

Education

Bachelor of Technology (B.Tech) in Computer Engineering

2022 - 2026

Pillai College of Engineering, University of Mumbai, Mumbai

CGPA : 7.7

Skills

Programming Languages: Java ,C ,Python, Javascript.

Databases: PostgreSQL , MySQL, Redis

Tools: Git, VS Code, IntelliJ IDEA, Jupyter Notebook

Frameworks: React, Node.js, Spring Boot

AI/ML: RAG, Fine-Tuning (LoRA), LangChain, Vector Databases (ChromaDB).

Operating Systems: Linux, Windows, macOS

Other: Data Structures & Algorithms, Object-Oriented Programming (OOP), RESTful API Development

Soft Skills : Team Collaboration, Leadership and Mentoring, Communication, Time Management, Problem-Solving, Adaptability

Projects

CodeEcoScan

Static analysis tool that estimates energy risk and carbon footprint of Python workloads. Scores code across loop complexity, recursion depth, import weight, and I/O patterns using a custom AST-based scoring engine.

- Architected a real-time carbon emissions estimator – parameterized by hardware profile (Laptop / Cloud VM / GPU Server), runtime, and run frequency – converting energy draw to kgCO₂/day.
- Deployed a live public API on Render; built a terminal-inspired React frontend with syntax-highlighted heatmap overlays and score breakdowns by category.

Agent Sentinel

Autonomous Misinformation Defense System Designed and built a fully autonomous AI agent that detects, verifies, and neutralizes viral misinformation in near real-time for Government (B2G) and Enterprise (B2B) use. *Role:* Lead Backend Engineer. Technology used:

Python, FastAPI, HuggingFace Transformers (RoBERTa), GDELT API

Changelog-AI

Developed an end-to-end NLP pipeline using Gemini models and LangChain to transform raw GitHub commit metadata into structured, user-centric release notes. Deployed a high-availability backend on Railway with full observability via Sentry, monitoring model performance and capturing edge-case parsing failures in real-time.

AI Tutor with Voice Cloning & Persona Preservation

A full-stack AI application featuring a custom-trained tutor that answers questions about specific course materials. Achieved a 95% response accuracy on course-specific questions while maintaining the tutor's unique persona. *Type:* Group(4 members). *Technology Used:* Python, FastAPI, Streamlit, RAG, LoRA Fine-Tuning, Voice Cloning (XTTS), Vector Databases (ChromaDB), MLX.

SmartDoc Query Engine

A full-stack, AI-powered RAG (Retrieval-Augmented Generation) system for secure PDF querying and management. Architected a microservice-based solution using Docker Compose for high availability, leveraging PostgreSQL with pgvector for efficient vectorized storage and Redis for caching and rate limiting. *Role:* Full-Stack Developer / DevOps. *Technology Used:* Python (FastAPI), React.js (Vite), PostgreSQL (pgvector), Redis, Docker, Sentence Transformers, RAG.

Health Food AI Assistant

Built an intelligent AI assistant to provide personalized health recommendations. Action: Implemented a custom algorithm that analyzes user medical conditions against a database of 500+ food items. Result: Delivered instant, safe food suggestions to improve user dietary choices. *Type:* Self. *Technology Used:* JavaScript, AI Integration, Healthcare Domain.

DevConnect Platform

A comprehensive campus placement and academic resource sharing platform. Developed both frontend (React) and backend (Spring Boot) components *Type:* Self. *Technology Used:* Java Spring Boot, React.js, Full-Stack Development.

Concurrent Web Crawler

Developed high-performance web crawler using concurrent programming. The project is a piece of test automation.Engineered a multithreaded design that successfully scanned 1,000+ links per minute, identifying all broken links on a target website. *Type :* Self. *Technology Used :* Java, Multithreading, Web Scraping.

Languages

English, Hindi, Marathi